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1. Matrix Form of Lorentz Transformation

a)

$$\mathbf{A}^T\eta\mathbf{B} = \begin{pmatrix} A_t & A_x \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} B_t \\ B_x \end{pmatrix}$$

$$= \begin{pmatrix} A_t & A_x \end{pmatrix} \begin{pmatrix} -B_t \\ B_x \end{pmatrix}$$

$$A^T\eta\mathbf{B} = -A_tB_t + A_xB_x$$

which is indeed equivalent
b)
We want to show that

$$(\mathbf{\Lambda A})^T\eta(\mathbf{\Lambda B}) = \mathbf{A}^T\eta\mathbf{B}$$

and the key idea is that

$$(\mathbf{\Lambda A})^T\eta(\mathbf{\Lambda B}) = \mathbf{A}^T(\mathbf{\Lambda}^T\eta\mathbf{\Lambda})\mathbf{B}$$

So it suffices to prove

$$\mathbf{\Lambda}^T\eta\mathbf{\Lambda} = \eta$$

The Lorentz matrix is

$$\mathbf{\Lambda} = \begin{pmatrix} \cosh\rho & -\sinh\rho \\ -\sinh\rho & \cosh\rho \end{pmatrix}$$

Note that it's symmetric which means its transpose is itself
Let's do this

$$\eta\mathbf{\Lambda} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \cosh\rho & -\sinh\rho \\ -\sinh\rho & \cosh\rho \end{pmatrix} = \begin{pmatrix} -\cosh\rho & \sinh\rho \\ -\sinh\rho & \cosh\rho \end{pmatrix}$$

Now let's multiply it by its transpose
Top left is

$$\cosh\rho(-\cosh\rho) + (-\sinh\rho)(-\sinh\rho) = -\cosh^2\rho + \sinh^2\rho = -1$$

Top right is

$$\cosh\rho(\sinh\rho) + (-\sinh\rho)(\cosh\rho) = 0$$

Bottom left is

$$(-\sinh\rho)(-\cosh\rho) + \cosh\rho(-\sinh\rho) = 0$$

Bottom right is

$$(-\sinh\rho)(\sinh\rho)+\cosh\rho(\cosh\rho)=-\sinh^2\rho+\cosh^2\rho=1$$

So

$$\mathbf{\Lambda}^T\eta\mathbf{\Lambda}=\begin{pmatrix}-1 & 0\\ 0 & 1\end{pmatrix}=\eta$$

2. Pauli Matrices

a)
To find the eigenvalues we use the equation

$$(\mathbf{M}-\lambda\mathbf{I})=0$$

and impose that the determinant must be zero
σ_x:

$$\begin{vmatrix}-\lambda & 1\\ 1 & -\lambda\end{vmatrix}=\lambda^2-1=0\to\lambda=\pm1$$

which has eigenvectors
$\lambda=+1$

$$\begin{pmatrix}1\\1\end{pmatrix}$$

$\lambda=-1$

$$\begin{pmatrix}1\\-1\end{pmatrix}$$

σ_y:

$$\begin{vmatrix}-\lambda & -i\\ i & -\lambda\end{vmatrix}=\lambda^2-1=0\to\lambda=\pm1$$

which has eigenvectors
$\lambda=+1$

$$\begin{pmatrix}1\\i\end{pmatrix}$$

$\lambda=-1$

$$\begin{pmatrix}1\\-i\end{pmatrix}$$

σ_z:

$$\begin{vmatrix}1-\lambda & 0\\ 0 & -1-\lambda\end{vmatrix}=1-\lambda^2=0\to\lambda=\pm1$$

which has eigenvectors
$\lambda=+1$

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$\lambda = -1$

$$\begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

b)

$$\mathbf{M} = \begin{pmatrix} C_0 + C_3 & C_1 - iC_2 \\ C_1 + iC_2 & C_0 - C_3 \end{pmatrix}$$

Thus, rewriting in terms of the matrix elements, we have

$$C_1 = \frac{\mathbf{M}_{1,2} + \mathbf{M}_{2,1}}{2}, C_0 = \frac{\mathbf{M}_{1,1} + \mathbf{M}_{2,2}}{2}, C_3 = \frac{\mathbf{M}_{1,1} - \mathbf{M}_{2,2}}{2}, C_2 = \frac{\mathbf{M}_{2,1} - \mathbf{M}_{1,2}}{2i}$$

c)
$\mathbf{M}^\dagger$ is

$$\mathbf{M}^\dagger = \begin{pmatrix} (C_0 + C_3)^\star & (C_1 + iC_2)^\star \\ (C_1 - iC_2)^\star & (C_0 - C_3)^\star \end{pmatrix} = \begin{pmatrix} C_0^\star + C_3^\star & C_1^\star - iC_2^\star \\ C_1^\star + iC_2^\star & C_0^\star - C_3^\star \end{pmatrix}$$

as this must be equal to $\mathbf{M}$, we have

$$C_0^\star + C_3^\star = C_0 + C_3, C_0^\star - C_3^\star = C_0 - C_3$$

which implies

$$C_0^\star = C_0, C_3^\star = C_3$$

and

$$C_1^\star - iC_2^\star = C_1 - iC_2, C_1^\star + iC_2^\star = C_1 + iC_2$$

which implies

$$C_1^\star = C_1, C_2^\star = C_2$$

this means

$$C_0, C_1, C_2, C_3 \in \mathbb{R}$$